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Examiner reports

Q1.

Most candidates obtained the method mark in part (a) by obtaining –0.32. Many candidates went on to complete this part correctly, although incorrectly writing their answer to 2dp was a common error. Answers in degrees were not common, but they were seen.

Part (b) was answered very well by the majority of candidates with the correct trigonometric identity being used.

In part (c) most candidates were able to successfully factorise the quadratic expression, and many went on to complete the question correctly. Marks were lost by 'extra' values being given or poor accuracy of writing their answers to 2dp.

Q2.

The vast majority of candidates were able to quote and use the correct formulae for the area of the sector and for the arc length, and most obtained both correct answers. Otherwise, the most common error in part (a) was to form the equation incorrectly, by writing "2 × 18 = $\frac{1}{2}r^2\theta$", before making a second error to obtain the printed answer. In part (b), a very small minority left the perimeter as 3 cm.

Q3.

In part (a), the vast majority of candidates quoted and used the sine rule correctly, but once again a significant number failed to show sufficient detail in their working to justify the printed answer to the degree of accuracy quoted. The examiners were looking for at least a calculated value of a relevant product or quotient, following a rearrangement, to be shown to two decimal places or more before the printed answer was stated.

EXAM PAPERS PRACTICE

The area of the triangle, required in part (b), was often found correctly, although some candidates used the much longer method of finding the length of AB and the length of the perpendicular from C, rather than finding angle C and using $\frac{1}{2}ab \sin C$.

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Q4.

Part (a) was well answered by the majority of candidates. Tan x = 0.5 and x = 0.46, 3.60 was a common error. The second angle was often incorrect. Few cases of angles in degrees were seen.

In part (b), most candidates used the correct identity and were successful in answering this part of the question. The main error was $\text{cosec}^2 x = \cot^2 x - 1$ followed by fudging of the rest of the solution.

In part (c), most candidates attempted to factorise the quadratic expression, although some used the quadratic formula. Those who factorised were usually correct, although solutions of ½ and –2 were not uncommon. Those who used the formula often made the

EXAM PAPERS PRACTICE

error of $\cot^2 x = 2$ or $\cot^2 x = -0.5$. Candidates with 3.60 as a solution in (a) were able to recover here but often failed to obtain both marks; 5.17 was a common error.

Q5.

This question, which tested radian measure, was not answered as well as expected. A significant number of candidates found the length of the chord PQ instead of the arc PQ in part (a).

In part (b), only a minority of candidates found the correct expression for α in terms of π.

Those using $\frac{3\pi}{7} + \alpha + \alpha = \pi$ usually went on to score both marks but it was not uncommon to see π replaced by 180 in candidates' initial statements. Those attempting to solve the problem by using the sine rule were rarely successful.

In the final part of the question a significant number of candidates found the perimeter of the sector instead of the perimeter of the shaded segment. It is worth recording that a significant minority of candidates who failed to score the marks in part (a) gave a correct expression for the arc length in part (c).

Q6.

Most candidates correctly used the cosine rule in part (a) but some failed to show sufficient detail in their working to be able to justify the printed answer to the degree of accuracy stated.

In part (b) most candidates found the correct answer for the area of the triangle although the incorrect expression $\frac{1}{2} \times 8.3 \times 8.56 \times \sin 65$ was seen more often than any other errors.

Part (c) was not answered well with many candidates assuming that the perpendicular was also the bisector of angle B. Those who used $\frac{1}{2} BC \times AD$ = answer (b) usually went on to score full marks in part (c).

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Q7.

In general this question was done well by candidates of all abilities. Part (a) was very well answered by the majority of candidates, although $\tan x = 1/3$ and $\sin x = 1/3$ were also seen. 1.23 was usually seen but the second result was often incorrect. There were very few cases of results given in degrees.

Most candidates used the correct identity part (b) in and were successful in answering this part of the question. The main error was candidates using $\tan^2 x = \sec^2 x + 1$ and then fudging the rest of the solution.

In part (c) most candidates attempted to factorise the result from part (b), although some used the quadratic formula. Those who factorised were usually correct, although solutions of $-1/3$ and 1 were not uncommon. For the final 3 answers many totally correct solutions were seen. Candidates with an incorrect solution in part (a) were able to recover here

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EXAM PAPERS PRACTICE

from follow through marks, but often candidates failed to obtain both marks since 0 or 6.18 often accompanied 3.14.

Q8.

The vast majority of candidates were able to quote and use the correct formulae for the area of the sector and for the arc length and most obtained both correct answers. Again, some weaker candidates quoted formulae from page 8 of the formulae booklet without understanding the meaning of them and gained no credit.

Q9.

Most candidates applied the cosine rule correctly, and convincingly obtained the printed result in part (a). Part (b) was beyond the capabilities of many candidates. A significant number tried to use the identity $\tan\theta = \frac{\sin\theta}{\cos\theta}$ without success. Those who quoted the correct identity, $\sin^2\theta + \cos^2\theta = 1$, gained a mark but it was disappointing to see some then take the square root of each term. In general only the better candidates reached the printed answer legitimately. It was pleasing to see many candidates recover to obtain the correct area of triangle ABC.

Q10.

Most candidates were able to give a correct value in part (a) but a significant minority did not give any second value, despite the hint in the question: "answer<u>s</u>". A common wrong answer was "4.407". In part (b)(i), some candidates mixed up the coordinates and in part (b)(ii) "($\pi + a$)" was the usual wrong answer. The description of the required transformation in part (c) was answered better this year.

Although some excellent solutions were seen to part (d), these were very much the exception. The vast majority of candidates could not deal correctly with the $2x$ beyond finding the first solution, $x = \frac{2\pi}{5}$.

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Q11.

Parts (a) and (b) were very well answered with most candidates scoring full marks. Surprisingly, part (c) proved to be demanding. Many candidates only found 2 rather than 4 solutions and, more alarmingly, candidates seemed to have a problem dealing with the (θ – 0.1) and often subtracted the 0.1, giving an answer of 0.63 rather than 0.83.

Q12.

This question proved to be a good source of marks for most candidates. A small minority of candidates failed to gain the marks in parts (a) and (b) because either they showed insufficient detail in their solution or they worked in degrees with some premature approximation at an earlier stage. Others quoted incorrect formulae; in particular $r^2\theta$ was sometimes stated for the area of the sector. The required method for finding the area of the shaded region in part (c)(i) was generally well understood. However, in a minority of cases full marks were not gained due to calculators being set in degree mode.

EXAM PAPERS PRACTICE

The final part, using the cosine rule, was usually well answered, although a few could not manipulate the arithmetic correctly, some used degree mode and, perhaps more alarmingly, some candidates thought the cosine rule (as stated in the question) involved the used of sin(1.4). Candidates should be aware that the cosine rule is in the AQA formulae booklet.

Q13.

For many candidates this was their best answered question. In part (a) weaker candidates seemed to be unaware of the formula $\frac{1}{2}ab\sin C$ for the area of the triangle but the most common reason for the loss of a mark was not showing a value for θ other than the printed value and hence not showing that the result was correct to three significant figures. It was disappointing to see some candidates quoting the cosine rule with sin θ instead of cos θ (candidates should be aware that the cosine rule is given in the formulae booklet) but in general this part of the question was answered very well. Most candidates were able to quote the correct formulae for arc length and sector area but some recalculated the area of the triangle, quite often not getting the value 20 as given in the question. Some candidates quoted and used the incorrect formula $\frac{1}{2}r^2(\theta - \sin\theta)$ to answer part (c)(ii).

Q14.

Most candidates were able to quote and use the correct formula for the area of the sector to obtain the printed answer for θ. In part (b), a higher proportion of candidates than last year realised that the perimeter of the sector included the two radii. Some weaker candidates quoted formulae from page 8 of the formulae booklet without understanding the meaning of them. For example, the 'd' in '$A=\frac{1}{2}\int r^2\,\mathrm{d}\theta$,' was given the value 10 (presumably the length of the diameter).

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Q15.

It was disappointing to find a significant minority of candidates not using the sine rule to answer part (a). Those candidates who used the sine rule frequently failed to gain the final mark because they did not indicate a more accurate value to justify why the angle was 23.2° correct to the nearest 0.1°. In part (b), many candidates were able to find the area of the triangle by quoting and using the formula 1/2 $ab\sin C$, although too many used the wrong angle, 23.2°, for . Those who used Area = 1/2 × base × height were less successful.

Q16.

With the cosine rule in the formulae booklet, candidates normally used the cosine rule rather than using "$\sin\frac{\theta}{2}=\frac{16}{24}$". Some weaker candidates applied the rule incorrectly as illustrated by $24^2 = 24 + 32^2 - 2 \times 24 \times 32\cos\theta$". Candidates need to be aware that in order to "show that θ = 1.46 correct to three significant figures", they should have supplied a value for θ to a greater degree of accuracy. The vast majority of candidates found the

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EXAM PAPERS PRACTICE

length of the arc and it was particularly pleasing to see weaker candidates use the printed answer from part (a) to answer part (b). It was surprising to find a significant number of candidates giving the answer for the area of triangle ABC rather than the area of sector ABC in part (c)(i) and then producing a fully correct solution in part (c)(ii). The formula, $\frac{1}{2}ab\sin C$, for the area of a triangle did not seem to be as well known as it might have been.

Q17.

Many candidates were able to find the area of the triangle, although not all quoted and used the formula $\frac{1}{2}ab\sin C$. Some candidates had their calculators set in the wrong mode; both radians and grads were used.

Such candidates, in this examination, were penalised by no more than one mark. In part (b) it was disappointing to see so many candidates failing to gain all three marks. It was not uncommon to see the cosine rule quoted with cos C replaced by sin C.

A very common error was to forget to take the square root and give the answer for AB as 6.47. It would have been advantageous for such candidates to realise that the side opposite the angle of 30° cannot be the largest side of the triangle.

Q18.

Most candidates realised that the arc length was 1.5r but a significant minority could not form or solve the equation $r + r + 1.5r = 56$. The area of the sector required in part (b) was answered well, although some candidates used the incorrect formula $\frac{1}{2}r^2(\theta - \sin\theta)$.

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